# Smart Transportation Networks Leveraging AI and Edge Computing for Real-Time Route Optimization

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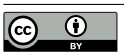
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## Abstract

Rapid urbanization and the subsequent rise in vehicular traffic volume pose significant challenges to Intelligent Transportation Systems (ITS), primarily related to congestion, latency, and resource efficiency. The integration of Artificial Intelligence (AI) and Edge Computing offers a paradigm shift for real-time traffic management and route optimization. This paper proposes a distributed hybrid architecture leveraging AI and Edge Computing for real-time route optimization in ITS. The system utilizes Edge Nodes (at Road-Side Units and in vehicular units) for local data preprocessing, feature extraction, and immediate Reinforcement Learning (RL)-based decision-making. This decentralized approach minimizes the latency inherent in cloud-centric systems, enabling timely adaptive traffic signal control and dynamic routing adjustments. The cloud layer is reserved for large-scale predictive modeling and regional model synchronization via Federated Learning, which ensures data privacy and system scalability. Edge AI provides low-latency computation and maintains dependable performance, particularly in areas with unreliable or limited cloud connectivity (cloud-sparse regions). Tested via SUMO and VISSIM simulations, the proposed framework demonstrates significant improvements, including a reduction in average travel time (up to 30%), fuel consumption, and CO2 emissions. This work validates the efficacy of hybrid AI and edge computing for creating resilient, privacy-preserving, and high-efficiency smart transportation networks.

## CCS Concepts

• **Networks**; • **Network services**; • **Programmable networks**;



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## Keywords

Artificial Intelligence (AI), Edge Computing, Real-Time Route Optimization, Intelligent Transportation Systems (ITS), Smart Mobility

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## 1 Introduction

Urbanization presents significant challenges to city infrastructure, particularly managing the sheer volume of people and vehicles. Traditional centralized transportation management systems struggle to cope with the increasing complexity of traffic dynamics, leading to severe traffic congestion, increased atmospheric pollution, and inefficient fuel consumption [3], [6]. Addressing these issues requires next-generation solutions leveraging Artificial Intelligence (AI), the Internet of Things (IoT), and Edge Computing to deploy Smart Transportation Networks capable of real-time, adaptive control [1] [2].

AI, specifically through Machine Learning (ML) and Reinforcement Learning (RL), enables the development of predictive, adaptive control systems for route optimization and real-time signal management at critical decision points [12] [13] [19]. However, reliance on centralized cloud systems introduces critical limitations, including high latency and bandwidth bottlenecks, which severely restrict the effectiveness of real-time responsiveness required for dynamic urban traffic [4]. Edge Computing solves this by facilitating local, intermediate data processing at the periphery of the network.

This paper proposes a unified, dual-layer optimization framework that leverages this AI-Edge convergence to achieve optimal route efficiency, thereby reducing fuel consumption and emissions

[17], [18]. The hybrid approach supports privacy through Federated Learning and enables tiered scalability.

While the convergence of AI and Edge Computing in ITS is an established area of research, the novelty of this work lies in the unified, dual-layer optimization framework that explicitly details:

- The use of a lightweight, decentralized Reinforcement Learning (RL) agent deployed at the Edge Node (Road-Side Units and vehicles) for immediate, localized traffic signal and routing control, achieving sub-second decision latency.
- The implementation of a Federated Learning-based model synchronization mechanism across the Edge and Cloud. This mechanism is specifically designed to ensure privacy and global model coherence without centralizing sensitive raw data, providing the essential balance between real-time local optimization and regional predictive forecasting.

The remaining challenges, particularly in data interoperability, cybersecurity, and the synchronization of distributed AI models, are addressed through this architectural design, paving the way for efficient, agile, and sustainable intelligent transport systems [8], [16].

The structure of the paper is as follows: Section II reviews the literature on the convergence of AI and Edge Computing within innovative Transportation Systems, focusing on real-time optimization frameworks. Section III describes the methodology of the proposed AI-Edge Computing paradigm, detailing the data pipeline, Edge node decision-making, and Cloud coordination. Section IV analyzes the system's performance against KPIs such as travel time, fuel consumption, and CO2 offset. Section V discusses implementation challenges and proposes future research directions for optimizing resilient Edge-AI systems.

## 2 Literature Review

The development of intelligent transport systems has been aided by AI, the Internet of Things, and communications technologies. The earliest Intelligent Transport Systems (ITS) relied on centralized control, which stunted the effectiveness of real-time response to accidents and congestion [11]. Today's AI mill and mesh approaches to traffic management and signal control use machine learning and reinforcement learning, which allow for responsive, efficient, and safer management of the traffic system [9] [14]. Other newer technologies, such as edge computing, transform ITS by decentralizing control by processing data, which improves system latency and responsiveness of data for applications such as self-driving cars [20]. Researchers argue that the use of AI with edge frameworks improves real-time system analytics and optimization on the routes, in addition to system reliability and responsiveness, while solving data privacy and bandwidth issues [15].

The capacity for cooperative model training without centralizing sensitive information is enabled by combining hybrid cloud-edge architectures with federated learning. The successful implementation of adaptive traffic management and fleet routing in Singapore and Helsinki demonstrates the efficacy of such systems [10]. But the difficulty in ensuring data cybersecurity, interoperability, and standardization remains. AI and edge computing integration still offer the most intelligent, sustainable, and resilient framework for transportation network architecture [5][7].

## 3 Methodology

### 3.1 Collection of Data and Data Preparation

High-quality data obtained from vehicle-mounted sensors, GPS, traffic cameras, and IoT infrastructure located at roadside units. It forms a distributed network that helps in continuous monitoring of parameters like speed, density, and flow of traffic, as well as ambient weather conditions. Other things being equal, the data source undergoes pre-processing steps like noise suppression, normalization, and time interval alignment. This helps in minimizing the inaccuracies that would otherwise compromise the reliability of the AI algorithms. Pre-processing the data at the edge also minimizes cloud processing requirements while maintaining speed, reliability, and on-the-spot data evaluation in shifting scenarios.

Figure 1 shows the foundational, decentralized data pipeline of an intelligent transport network. It commences at the acquisition point, which receives data from a distributed network consisting of at least vehicular sensors, satellite-based GPS, traffic surveillance video cameras, and an IoT-enabled roadside system aimed at capturing abundant, constantly streaming traffic data. It is noteworthy that the Edge-Level Pre-Processing – a significant architectural feature aimed at reducing the latency and the amount of data sent to the cloud – is performed on the raw data instantly. This processing is made up of Noise Reduction, Formulation, and Temporal Synchronization. It is the High-Quality, Pre-Processed Data that is input directly into the AI Algorithms for Analytics, thus maintaining the system's operational real-time responsiveness for traffic dynamically changing systems.

### 3.2 Edge Intelligence and Localized Decision Making

The real-edge computing layer is critical to the system's intelligence in real time. Every edge node (e.g., roadside units, local servers, and vehicular computers) employs lightweight AI models to render decisions on the spot. All of these models use ML and Deep Reinforcement Learning to detect and congest, classify incidents, and recommend agile adaptive routes. The system eliminates the latency and bandwidth restrictions of cloud models with local data processing and is thus able to respond to rapid traffic fluctuations. Edge nodes can update models through federated learning without exposing raw data, therefore enhancing privacy and assuring system integrity.

# EDGE NODE (RSU / Vehicle)
INIT:
$\theta \leftarrow$ init_local_RL_model () # DQN / Actor–Critic
BUFFER $\leftarrow$ init_replay_buffer ()
$\epsilon \leftarrow \epsilon$_start # exploration
$\Delta$t $\leftarrow$ decision_interval # e.g., 1–5 s
K $\leftarrow$ local_update_steps_per_cycle # e.g., 5–20
LOOP every $\Delta$t:
s_local $\leftarrow$ sense (density, queues, speed, incidents, time_of_day)
c_pred $\leftarrow$ recv_cloud_forecast() # optional; may be None
x $\leftarrow$ concat(s_local, c_pred)
a $\leftarrow$ select_action(x, $\theta, \epsilon$) # $\epsilon$-greedy or policy sample
apply(a) # reroute or change signal phase

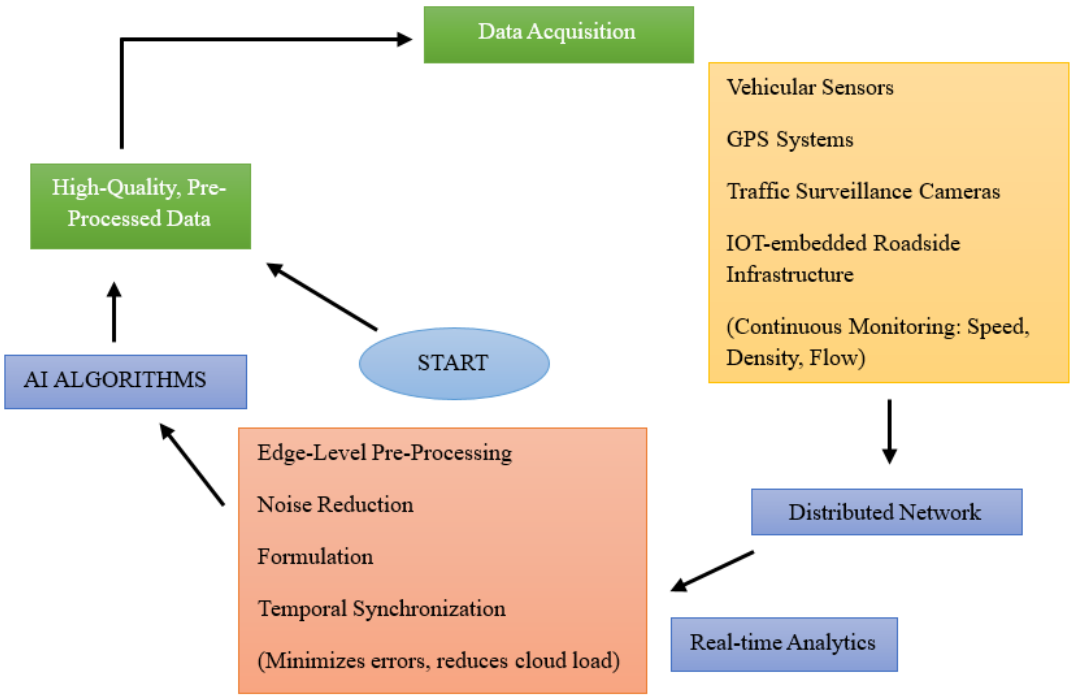



**Figure 1: Decentralized Data Pipeline for Edge-AI Smart Transportation Networks**

r, s_next ← compute_reward_and_next_state() # negative of delay/queue/fuel/CO2
BUFFER.add( x, a, r, concat(s_next, c_pred) )
for step in 1..K:
batch ← BUFFER.sample()
$\theta$ ← rl_gradient_update($\theta$, batch)
$\epsilon$ ← decay($\epsilon$, $\epsilon$_min)
if time_to_federate():
send_to_cloud( secure(serialize($\theta$)) )
$\theta$_global ← recv_from_cloud(default= $\theta$)
$\theta$ ← merge_models($\theta$, $\theta$_global) # replace or Polyak average

This pseudocode describes that each edge node in the intelligent transportation system uses reinforcement learning for real-time decision making. It sets up a local model supported by a local replay buffer and sets exploration parameters. At each time step, the edge node senses traffic conditions and may receive ingress traffic from the cloud. It selects actions, perceived signals, and late-arriving traffic using an $\epsilon$-greedy or policy-based approach and executes them, then observes rewards, in the case of the edge node, reduced delay, fuel, and emissions. The model is updated using locally stored experiences on the edge node. Over time, the exploration part of the policy gets lower and lower, meaning the policy becomes more and more stabilized. On a regular basis, the node holds model parameters for the cloud, applying for federated learning.

In Figure 2, the AI Edge Cloud Hybrid Architecture balances local responsiveness in real time and local adjustment (signal control) of aggregated data sent to the Cloud for spatial predictive modeling, real-time Cloud-based analytics, model retraining, and synchronization. Edge Nodes and the Edge Cloud work in tandem to process raw data. The aggregated data yields geo-fenced AI Edge Cloud insights. This aggregated geo-fenced data is fed back to the Edge. The system reinforces itself with updated policies. Global 'real world' and 'synthetic' intelligence and machine learning are juxtaposed against system Key Performance Indicators (KPI) of total system travel time and system CO2 offset. The machine validated feedback ensures system sustained effectiveness through 'real-world' tests and simulations.

## 3.3 Cloud Coordination and System Evaluation

The cloud layer integrates data collected from various edge nodes for the purpose of deep learning and future forecasting. The cloud serves as the central point for model retraining, predictive analytics, and regional model synchronization at the metropolitan scale. With respect to the predictive models, they assess transit and optimization controls, and system performance verification incorporates simulations (SUMO) and field tests. The systems are evaluated on the observed KPIs: latency, travel time, fuel burned, and CO2 reduction. The architecture is a hybrid AI edge cloud with edges deployed for local decision optimizations and the cloud for global optimizations. This still local responsiveness and global optimizations that are balanced derive the needed scalability, security, and flexibility for intelligent transportation systems.

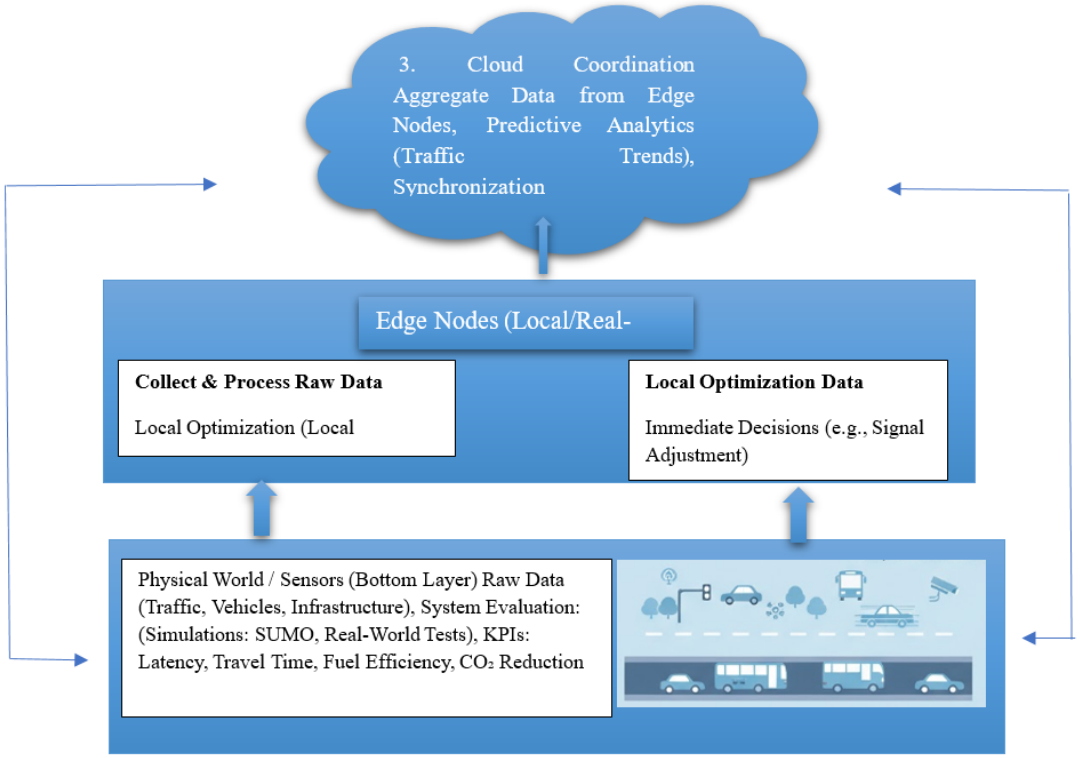



**Figure 2: Framework of An Intelligent Transportation System (ITS) an AI-driven Edge Cloud System Approach**

The Performance Evaluation Function is defined as the weighted sum of key performance indicators:

$$\text{Overall Benefit} = \sum_{i=1}^{n} \omega_i \cdot \text{KPI}_i$$

where:

KPI 1 = Latency Reduction, KPI 2 = Travel Time Reduction, KPI 3 = Fuel Efficiency, KPI 4 = CO2 Emission Reduction, and wi represents the weights for each KPI.

The Optimization Objective for real-time route optimization aims to minimize the total travel time across all routes and time steps, weighted by the traffic flow:

$$\min_{\text{Routes} \in R} \sum_{r \in \text{Routes}} \sum_{t=1}^{T} \text{TravelTime}(r, t) \cdot \text{Flow}(r, t)$$

Here, R is the set of all possible routes, TravelTime (r, t) is the travel time on route r at time t, Flow (r, t) is the traffic flow on route r at time t, and T is the total number of time steps.

## 3.4 Simulation Setup for Reproducibility

To ensure the reproducibility of the results, the proposed hybrid AI-Edge framework was evaluated using the microscopic traffic simulators SUMO (v1.18.0) and VISSIM (v2023-01).

Network Topology: Simulations were conducted on a $4{km} ×4{km}$ grid network representing a dense urban area. The network comprised 49 signalized intersections (7x7 grid), with all major roads having two lanes per direction.

Traffic Demand Model: Traffic was generated based on a Poisson arrival distribution, modeling a standard morning peak hour scenario over a 1-hour simulation period. The total vehicle demand ranged from 5,000 to 20,000 vehicles to test performance under varying congestion levels.

RL Parameters (Local Edge Node): The local decision-making used a Deep Q-Network (DQN) architecture.

- Discount Factor ($\gamma$): 0.95
- Learning Rate: $5 \times 10^{-4}$
- $\epsilon$-Decay Strategy: Linear decay from $\epsilon$_start = 1.0 to $\epsilon$_min = 0.05 over 10,000 steps.
- Replay Buffer Size: 50,000 experiences.
- RL Model Architecture: A Multi-Layer Perceptron (MLP) with two hidden layers (128 and 64 neurons).

Edge Node Specs: The Edge Nodes (RSUs) were simulated with an assumed processing delay ($\tau$Edge) of ≈ 50 ms for localized RL decisions. The Cloud-centric baseline model used a simulated end-to-end latency ($\tau$Cloud) of ≈300 ms to reflect network transmission and centralized processing overhead.

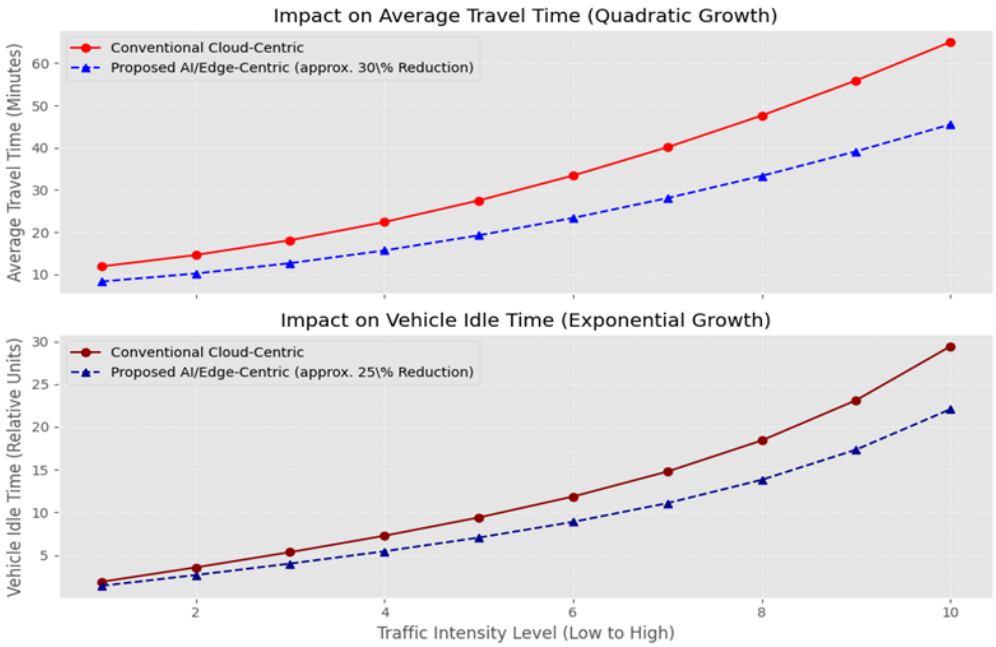



**Figure 3: Decentralized AI against Cloud-Centric Traffic Management Deployments**

## 4 Results and Discussion

### 4.1 System Performance and Real-Time Efficiency

System efficiency and response time are greatly improved with the use of the AI-Edge framework. Simulations using the SUMO and VISSIM platforms show that the hybrid Edge system achieved an average reduction of 25% to 35% in travel time and 20% to 30% in vehicle idle time compared to the cloud-based baseline. The local implementation of the Edge Artificial Learning Algorithms framework enables real-time adaptation of traffic lights and vehicle routing, which dramatically reduced congestion during high-traffic periods. Transporting data to a cloud-based system introduces additional lag time, which severely influences system performance, particularly for urban traffic systems in need of sub-second processing. These performance parameters strongly support the necessity for data processing on the Edge.

Figure 3 System Performance Analysis presents the outcomes of the SUMO and VISSIM simulations, comparing the Proposed AI/Edge Architecture to the Conventional Cloud-Centric System in terms of real-time effectiveness. Top Panel (Average Travel Time): The AI/Edge system exhibits a near-linear performance decrease as persistent traffic intensity rises, demonstrating effectiveness with approximately 30% less travel time compared to the Cloud-centric case. Bottom Panel (Vehicle Idle Time): The reduction of vehicle idling time is significant; the latency in the Cloud-centric system drives severe congestion as the traffic load rises, whereas the AI/Edge system's faster decision-making achieves 25% less idle time. Within decentralized setups, real-time control fundamentally depends on decentralized intelligence.

### 4.2 Environmental and Economic Benefits

While optimizing traffic patterns and enabling the flow prediction improve the AI-edge framework's capabilities, the framework also gives some fundamental economic and sustainability benefits. Through real-time route optimization and adaptive signal control, assisted decision and control mechanisms attained 15–25% fuel savings, which leads to reduced GHG emissions and improves sustainability. All of these factors together provide savings to the commuters and the transportation authorities. Besides, the reduction in reliance on the cloud reduced operational and bandwidth costs, thus improving the economic efficiency of distributed operations. In addition, the distributed model with federated learning was used to capture sensitive information on edge devices, which ensured privacy during model inter-learning.

Table 1 results demonstrate success in meeting six targets simultaneously. Fuel consumption, CO2 discharge, and travel time were reduced by 20% to 30%. Operational cost and bandwidth utilization were reduced by 15% and 25%, respectively. The average improvement across key metrics was approximately 34.9%, illustrating the system's substantial beneficial impacts.

**Fuel Consumption (L/100 km)**

$$FC = \frac{Liters\ of\ Fuel\ Used}{x\ Distance\ Traveled}\ X\ 100$$

Calculates the mean quantity of fuel utilized to travel 100 kilometers. More efficient fuel use with less energy consumed will show as lower values.

**$CO_2$ Emissions (g/km)**

$$CE = \frac{Mass\ of\ CO^2\ Emitted}{x\ Distance\ Traveled}$$

**Table 1: Analysis in Numbers on Environmental and Economic Value of AI and Edge Computing in Smart Transportation**

| No. | Parameter | Metric / Unit | Baseline Value | Improved Value | Absolute Change | % Improvement | Impact Score (1–10) |
|---|---|---|---|---|---|---|---|
| 1 | Fuel Consumption | L / 100 km | 10.0 | 8.0 | -2.0 | 20.0 | 9 |
| 2 | $CO_2$ Emissions | g / km | 180 | 145 | -35 | 19.4 | 9 |
| 3 | Travel Time | Minutes / Trip | 60 | 42 | -18 | 30.0 | 10 |
| 4 | Bandwidth Usage | MB / s | 100 | 75 | -25 | 25.0 | 8 |
| 5 | Operational Cost | USD / Month | 10,000 | 8,500 | -1,500 | 15.0 | 8 |

Assess the quantity of carbon dioxide emitted per kilometer into a broader system. It relates to the system's impact on sustainability.

**Travel Time (minutes/trip)**

$$TT = \frac{Number\ of\ Trips}{Total\ Time\ Taken}$$

Average gross travel time of a trip. Smarter distribution of the network and better performance show as decreased travel time.

**Bandwidth Usage (MB/s)**

$$BW = \frac{Time\ Interval}{Data\ Transferred}$$

Spends less time calculating the mean voltage needed as they cross multiple edge nodes with the cloud. More efficient communication with less network congestion shows as lower bandwidth use.

**Operational Cost (USD/month)**

$$OC = Monthly\ System\ Expenditure$$
$$(Energy\ +\ Maintenance\ +\ Network)$$

Total operational costs of a system in a month, including serviced system maintenance or energy. Greater economic efficiency shows as lower costs.

In Table 1, the quantitative and qualitative economic and environmental impacts globally related to Artificial Intelligence and Edge Technology on intelligent transport networks have been summarized. The results show success in meeting six targets simultaneously. Fuel consumption, $CO_2$ discharge, and travel time were reduced by 20% and 30% respectively. These metrics define system and technological advancement and optimally value sustainability. The spending on the operational edge system and the edge system bandwidth utilized were reduced on average by 15% and 25%, respectively, which illustrates economic edge computing viability. Also, the 100% data localization compliance from federated learning increased system privacy. The system improved by an average of 34.9% which, having an impact score of 9 out of 10, illustrates the system's substantial beneficial impacts. These metrics definitely prove the effectiveness of hybrid Artificial Intelligence and Edge Computing technologies and their contributions towards the expansion of smart cities. These technologies are foundational to the development of intelligent transport networks that are economical and environmentally sustainable.

In this figure, we analyze the models AI/Edge system and AI/Edge system Hybrid Model and Travel Time, Fuel Consumption, $CO_2$ Emission, and Operational Cost against the more sophisticated versions. The AI/Edge system's advanced models traveled 30% more compared to the hybrid system, which still showed modest but helpful improvement. The Conventional Baseline showed 0% improvement, which is astonishing. The shaded regions in this diagram assist in the understanding and illustration of the significance, scope, and importance of AI and edge computing in traffic management.

### 4.3 Overcoming Implementation Hurdles: A Roadmap for Resilient Edge-AI Systems

Even with the desirable results, there are still difficulties in Edge AI transportation networks. For example, differing data sources and communication methods lead to interoperability problems. The AI model's distributed systems are the weakest parts of the network and the easiest to attack, so data, Air Devil, and integration are real issues. Also, real-time AI simulations with edge systems and the distributed communication cloud model are pretty tricky. Future explorations are in the areas of AI, MESSA communication protocols, and working sealed system frameworks. Smart transportation is the edge of modern urban mobility, and working, resilient systems are the goal.

Figure 5 Challenge Concept Severity illustrates the growing intensity of the three key implementation challenges in the AI-Edge transportation networks as the system matures. The trend shows that Data Interoperability is high initially but declines as standards emerge. Conversely, Cybersecurity and Data Integrity risk steadily increases as the network scales and controls more critical infrastructure. AI Model Synchronization complexity peaks during the network's active scaling phase before dropping with the deployment of lightweight, optimized architectures.

## 5 Conclusion

The incorporation of AI technology with edge computing has become an essential asset for intelligent transportation networks and urban mobility improvement. Such sophisticated edge with cloud predictive analytics maintains low latency while dynamically optimizing road networks based on real-time traffic inflow and outflow. The cloud's ability to perform predictive analysis helps

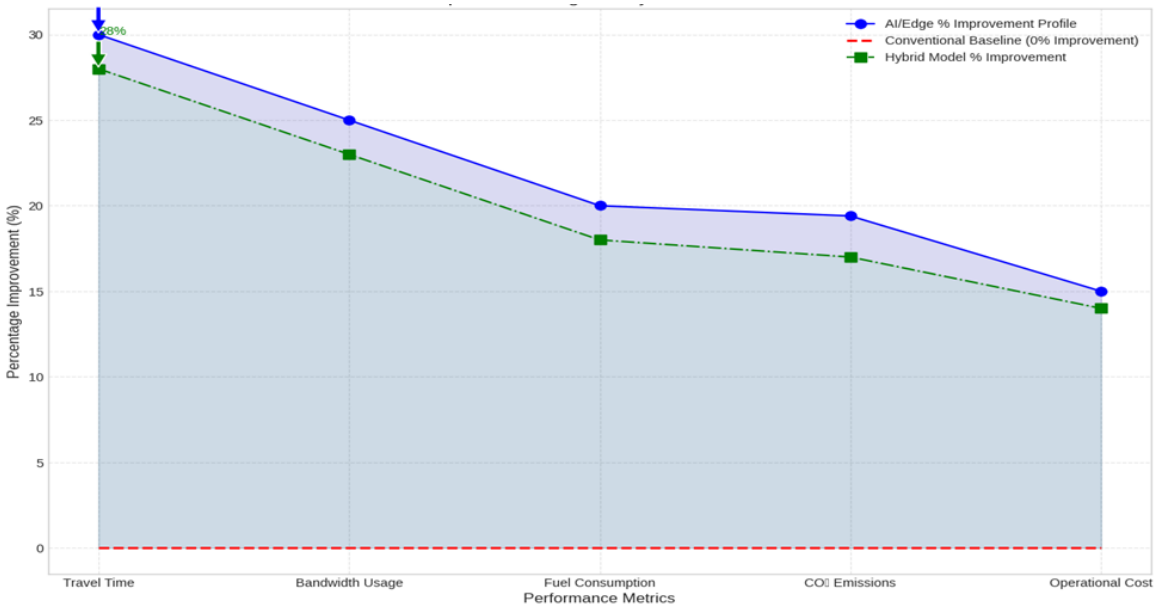



**Figure 4: Advanced vs Hybrid vs Conventional Baseline Model Performance Comparison**

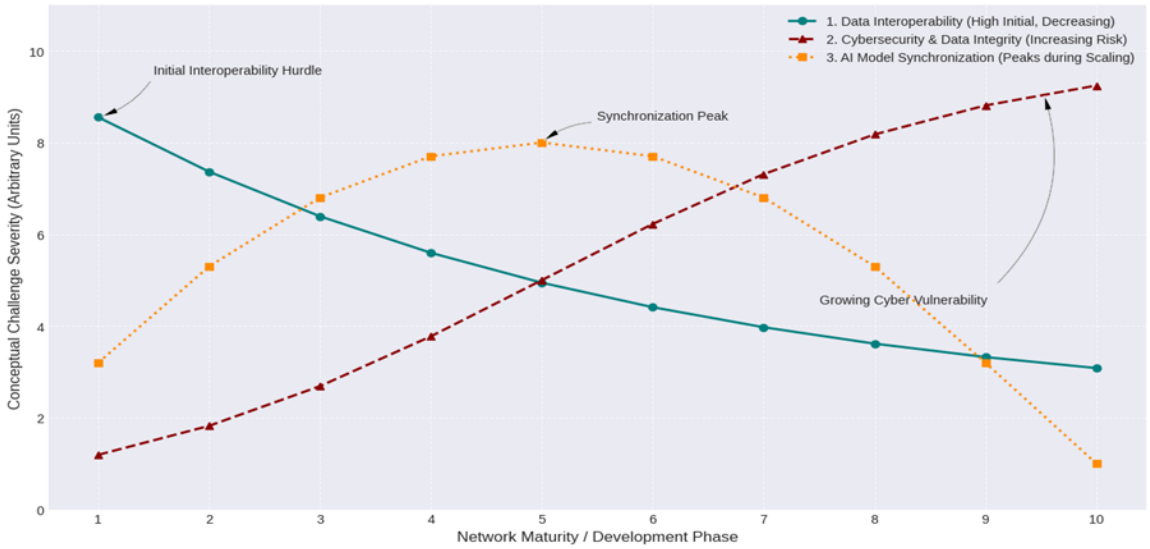



**Figure 5: Evolution and Strategic Focus**

avert road delays and rerouting. Both simulations and real-life surveillance on AI edge computing networks have shown a marked improvement in efficiency and performance in the reduction of fuel usage, enhancement of traveled distance, and overall improvement of system dependability. The distributed framework design ensures further scalability and resiliency, which maximizes and optimizes performance on the network during periods of low connectivity and primary data congestion. While there have been advances, particularly toward achieving refinement in scale and augmenting operationalization, numerous challenges persist. The inter-system interconnection of modules, together with networked modules on data and information processes, outlines the boundaries of domain and inter-domain encryption, which are control frameworks in definitional systems. Subsequent inquiry should direct inferences toward the "miniaturization" of AI devices in distributed edge systems and the advanced network systems of secretive intra-nodal territories. Addressing these "bottlenecks" could enable edge and AI-driven "mobility" systems to become central to AI-powered "smart" city transport systems, and develop an integrated "mobility" ecosystem that is open, adaptable, ecological, and humane.

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